

landing foot away from the ball. Then, because of that, they pull their other foot away from the ball (a drop step). So, they have compounded the flaw of being late with the flaw of moving away from the ball.

Another flaw can be blamed on the belief that landing on the push-off leg stores significant elastic energy. This is not the case. The amount of energy stored by the split step — or any other pre-jump — is primarily due to muscle stretching and activation in the legs rather than elastic energy. Elastic energy only increases the height of jumps and lunges by 5 per cent to 10 per cent. The extra height of countermovement jumps (jumps after a pre-jump) can be increased much more by increasing the amount of knee bend achieved after the split step — that is, the pre-hop knee bend plus the after-landing knee bend — than by landing more forcefully on one foot.

The knee bend that produces the most force of a jump, or lunge, is roughly 90°, and this number varies from player to player. It can also be achieved in both legs by landing simultaneously on both feet. But players are not doing this. Few push off for their first move to the ball with that much knee bend.

However, players could land their split step on both feet simultaneously at the instant their opponent hits the ball. If the sum of their knee bend when they hopped into the split step and the additional after-landing knee bend was roughly 90°, they could maximise the speed of their first move to the ball. If the duration of their after-landing knee bend was 0.2 seconds, they could make their first move to the ball with a powerful lunge 0.2 seconds after their opponent hit the ball.

To their great detriment, no high-level players are doing that.

How does a teaching pro or coach correct these flaws?

With considerable difficulty. The timing of the split step is learned at an early age and repeated thousands of times as a player progresses to high-level

status. It becomes a habitual, unconscious act. Changing its timing involves focusing on the change. This is possible during practice, but it may be fatal during matches that count. During those matches, a focus on anything except the ball and the opposing player can be a crippling distraction. So, changing the split step is best done by purposeful practice, including practice matches, rather than by trying to remember to do it during matches that count.

However, there are three situations where a changed split step can be advantageous even if not habitual. The first is returning serve. Players have time to think before returning serve. They could focus on hopping with bent knees and landing their hop at the instant the server hits the serve. They even have time to rehearse this changed split step.

The second is when approaching the net. It is not difficult to remember

Taking a moment: Among the situations where a changed split step can be advantageous even if not habitual is when returning serve, when players can focus on hopping with bent knees and land at the instant the server hits the serve.

that your first volley is likely to be challenging. This can be a trigger to use the changed split step. The third is similar to the second. When you see that your opponent is setting up for what could be an unusually challenging shot, that can also be a trigger to use the changed split step.

Should you do split steps differently on each of the three Major surfaces?

The slick surfaces of grass and clay can make landing the split step with a wide stance precarious. The wider the stance and the slicker the surface, the greater the tendency for the feet to slide away from each other on the split step touchdown. Therefore, players should experiment with stances of narrower width on grass and clay. But the touchdown should still be at the instant the opponent hits the ball.

Is the split step used in other sports?

In football, goalkeepers do split steps to block shots and penalty kicks. Baseball infielders do it when batters contact the ball. Cricket players do it when the ball hits the bat.

The split step is also used in squash, badminton, padel, U.S. football, rugby, volleyball, and boxing. In these sports, coaches teach, and athletes perform their versions of split steps. They understand the great importance of balance, quick reaction times, and explosive starts to get to the ball on time.

Performing a split step is spoken of as an intuitive act, the specifics of which are unconsciously developed by elite athletes.

